FOURIER [33. - 65.]

~ Red, transformacija, integral ~

33. Dirichletove uvjete na intervalu $[-1,1]$ ne zadovoljava

(B) $f(x)=\frac{1}{x^2}$ ima prekid u $x=0$ ✓

34. Zadovoljava Dirichletove $[-1,1]$

(B) $f(x)=x\cdot\sin x$ ✓

35. —||— (A) $x\cdot\sin x$ ✓

36. —||— (B) $f(x)=\frac{\sin x}{x}$ + ljeti se limesa ✓

37. Fourierov red: $f(x)=-3(1-\cos(2x))$, $x\in\left[-\frac{\pi}{4},\frac{\pi}{4}\right]$, $T=\frac{1}{2}\pi$

$a_0=?$

$$S(x)=\frac{a_0}{2}+\sum_{n=1}^{\infty} a_n\cos\left(\frac{2\pi n x}{T}\right)+b_n\sin\left(\frac{2\pi n x}{T}\right)$$

$$a_0=\frac{2}{T}\int_a^b f(x)\,dx=\frac{4}{\pi}\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}}-3(1-\cos(2x))\,dx=\frac{4}{\pi}\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}}-3+3\cos(2x)\,dx$$

$$=\frac{4}{\pi}\left(-3x+\frac{3}{2}\sin(2x)\right)\Big|_{-\frac{\pi}{4}}^{\frac{\pi}{4}}=\frac{4}{\pi}\left(-3\cdot\frac{\pi}{2}+\frac{3}{\cancel{2}}\cdot\cancel{2}\cdot 1\right)=\boxed{\frac{-6\pi+12}{\pi}}$$ (B) ✓

38. Fourierov red: $\underbrace{f(x)=2-|x|}_{\text{parna fja}}$, $x\in[-1,1]$, $T=2$, $a_0=?$

$$a_0=\cancel{2}\cdot\frac{2}{\cancel{2}}\int_0^1 2-x\,dx=2\int_0^1 2-x\,dx=2\left(2x-\frac{1}{2}x^2\right)\Big|_0^1=2\left(2-\frac{1}{2}\right)=\boxed{3}$$ (C) ✓

39. Fourierov red: $f(x)=-6\sin^4 x$, $a_0=?$

$$a_0=\frac{2}{T}\int_a^b -6\cdot\sin^4 x\,dx$$

$$\sin^2 x=\frac{1-\cos 2x}{2}\Big/^2$$

$$\sin^4 x=\frac{(1-\cos 2x)^2}{4}=\frac{1-2\cos 2x+\cos^2 2x}{4}$$

$$a_0=\frac{2}{T}\int_a^b -6\cdot\frac{1-2\cos 2x+\cos^2 2x}{4}\,dx$$

$$\cos^2 2x-\sin^2 2x=\cos 4x$$

$$a_0=\frac{-3}{T}\int_a^b 1-2\cos 2x+\frac{1}{2}\cos 4x+\frac{1}{2}\,dx \qquad \cos^2 2x=\frac{\cos 4x+1}{2}$$

$$a_0=\frac{-3}{T}\int_a^b \frac{3}{2}-2\cos 2x+\frac{1}{2}\cos 4x\,dx$$

T za kosinus $=[0,2\pi]$

$$=\frac{-3}{2\pi}\left[\frac{3}{2}-\cancel{\sin 2x}+\cancel{\frac{1}{8}\sin 4x}\right]\Big|_0^{2\pi}=\frac{-3}{\cancel{2\pi}}\cdot\frac{3}{2}\cdot\cancel{2\pi}=\boxed{\frac{-9}{2}}$$ (D) ✓

40. Fourier red: $f(x) = 4\sin^4 x$, $a_0 = ?$

$$a_0 = \frac{2}{T}\int_a^b 4\sin^4 x\, dx = \frac{8}{T}\int_a^b \frac{1}{4}\left(1 - 2\cos 2x + \cos^2 2x\right) dx$$

$$= \frac{2}{T}\int_a^b 1 - 2\cos 2x + \frac{1}{2}\cos 4x + \frac{1}{2}\, dx = \frac{2}{T}\int_a^b \frac{3}{2} - 2\cos 2x + \frac{1}{2}\cos 4x\, dx$$

$$= \frac{1}{\pi}\left(\frac{3}{2} - \sin 2x + \frac{1}{8}\sin 4x\right)\Big|_0^{2\pi} = \frac{3\cdot \cancel{2\pi}}{\cancel{2\pi}} = \boxed{3}$$

c ✓

41. Fourierov red: $f(x) = 3\sin^4 x$, $a_0 = ?$

$$a_0 = \frac{2}{T}\int_a^b 3\sin^4 x\, dx = \frac{6}{2\pi}\int_0^{2\pi} \frac{1}{4}\left(1 - 2\cos 2x + \cos^2 2x\right) dx = \frac{3}{4\pi}\int_0^{2\pi} \frac{3}{2} - 2\cos 2x + \frac{1}{2}\cos 4x\, dx$$

$$a_0 = \frac{3}{\cancel{4\pi}_{2}}\left(\frac{3}{2}\cdot 2\pi - \emptyset + \emptyset\right) = \boxed{\frac{9}{4}}$$

c ✓

42. Fourierov red $f(x) = -7(\sin^4 x - \cos^4 x)$, suma koeficijenata?

$$S(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\left(\frac{2\pi n x}{T}\right) + b_n \sin\left(\frac{2\pi n x}{T}\right)$$

$$f(x) = -7\underbrace{(\sin^2 x + \cos^2 x)}_{1}(\sin^2 x - \cos^2 x) = 7(\cos^2 x - \sin^2 x) = 7\cos 2x$$

* suma → samo a_n i b_n dio, odnosno s obzirom da je $7\cos 2x$ parna funkcija → a_n samo

$$a_n = \frac{2}{T}\int_a^b f(x)\cdot\cos(nx)\, dx$$

, odnosno $f(x) = 7\cos \underset{n=2}{2x}$ već je Fourierov red

$$a_1 = \frac{1}{\pi}\int_0^{2\pi} 7\cdot\cos 2x\cdot\cos x\, dx$$

↳ uz $\cos 2x$ → $a_2 = 7$

$$= \frac{7}{2\pi}\int_0^{2\pi} \underbrace{\cos 3x + \cos x\, dx}_{0}$$

koeficijenti su nula osim ako integral uspjemo promijeniti u sinus ili $\cos(x)$ + konst.

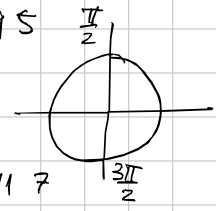
$$a_2 = \frac{7}{\pi}\int_0^{2\pi} \cos^2 2x\, dx = \frac{7}{2\pi}\int_0^{2\pi} \underbrace{\cos 4x}_{\emptyset} + 1\, dx = \frac{7\cdot \cancel{2\pi}}{\cancel{2\pi}} = \boxed{7}$$ a to je a_2

c ✓

43. Fourierov red: $f(x) = -3\left(1-\cos(3x)\right)$, $x \in \left[-\frac{\pi}{6}, \frac{\pi}{6}\right]$, $T = \frac{1}{3}\pi$, $a_5 = ?$

$$a_5 = \frac{6}{\pi}\int_{-\frac{\pi}{6}}^{\frac{\pi}{6}} \left(-3+3\cos 3x\right)\cos\left(\frac{2\pi\cdot 5\cdot x}{\frac{1}{3}\pi}\right)dx = \frac{-18}{\pi}\int_{-\frac{\pi}{6}}^{\frac{\pi}{6}} (1-\cos 3x)\cos(30x)\,dx$$

$$a_5 = \frac{-18}{\pi}\int_{-\frac{\pi}{6}}^{\pi/6} \cos(30x) - \frac{\cos 33x}{2} - \frac{\cos(27x)}{2}\,dx$$



$$a_5 = \frac{-18}{\pi}\left[\frac{1}{30}\sin(30x) - \frac{1}{66}\sin(33x) - \frac{1}{54}\sin(27x)\right]\Bigg|_{-\frac{\pi}{6}}^{\frac{\pi}{6}}$$

$$a_5 = \frac{-6}{\pi}\left[\underbrace{\cancel{\frac{1}{10}\left(\sin(5\pi)+\sin(5\pi)\right)}}_{\emptyset} - \frac{1}{\cancel{22}_{11}}\underbrace{\left(\sin\left(\frac{11\pi}{2}\right)\cdot\cancel{2}\right)}_{-1} - \frac{1}{\cancel{18}_{9}}\underbrace{\sin\left(\frac{9\pi}{2}\right)\cdot\cancel{2}}_{1}\right]$$

$$a_5 = \frac{-6}{\pi}\left[\frac{+1}{11} - \frac{1}{9}\right] = \frac{6}{\pi}\left[\frac{1}{9} - \frac{1}{11}\right] = \frac{\cancel{6}^{2}}{\pi}\cdot\frac{2}{\cancel{99}_{33}} = \boxed{\frac{4}{33\pi}}$$

B ✓

44. $f(x) = -\left(1-\cos(3x)\right)$, $x \in \left[\frac{-\pi}{6}, \frac{\pi}{6}\right]$, $T = \frac{1}{3}\pi$, $a_4 = ?$

$$a_4 = \frac{-6}{\pi}\int_{-\frac{\pi}{6}}^{\frac{\pi}{6}} \left(1-\cos(3x)\right)\cos\left(\frac{2\cancel{\pi}\cdot 4\cdot x\cdot 3}{\cancel{\pi}}\right)dx = \frac{-6}{\pi}\int_{-\frac{\pi}{6}}^{\frac{\pi}{6}} \cos(24x) - \cos(27x) - \cos(21x)\,dx$$

$$= \frac{-6}{\pi}\left[\frac{1}{24}\sin(24x) - \frac{1}{54}\sin(27x) - \frac{1}{42}\sin(21x)\right]\Bigg|_{-\frac{\pi}{6}}^{\pi/6}$$

$$= \frac{-1}{\pi}\left[\underbrace{\cancel{\frac{1}{4}\cdot 2\cdot\sin(4\pi)}}_{\emptyset} - \frac{1}{9}\cdot\underbrace{2\sin\left(\frac{9\pi}{2}\right)}_{1} - \frac{1}{7}\underbrace{\sin\left(\frac{7\pi}{2}\right)}_{-1}\right] = \frac{-2}{\pi}\left(\frac{1}{7}-\frac{1}{9}\right) = \boxed{\frac{-4}{63\pi}}$$

A ✓

45. Fourierov red: $f(x) = 2-|x|$, $x \in [-1, 1]$, perioda $T = 2$, $a_7 = ?$

$$a_7 = *\text{parna}* = \frac{4}{2}\int_0^1 (2-x)\cdot\cos\left(\frac{2\pi\cdot 7\cdot x}{2}\right)dx = 2\int_0^1 (2-x)\cos(7\pi x)\,dx$$

$$= 2\int_0^1 2\cos(7\pi x) - x\cos(7\pi x)\,dx = 4\cdot\frac{1}{7\pi}\left(\sin(7\pi x)\right)\Big|_0^1 = \emptyset$$

$$-2\int_0^1 x\,\frac{e^{-7\pi x}+e^{7\pi x}}{2}\,dx = -\int_0^1 x e^{-7\pi x} + x e^{7\pi x}\,dx$$

$u = x \rightarrow du = dx$

$dv = e^{-7\pi x} \rightarrow v = \frac{-1}{7\pi}e^{-7\pi x}$

$$-\left(\frac{-x}{7\pi}e^{-7\pi x}\Big|_0^1 - \frac{1}{49\pi^2}e^{-7\pi x}\Big|_0^1 + \frac{x}{7\pi}e^{7\pi x}\Big|_0^1 - \frac{1}{49\pi^2}e^{7\pi x}\Big|_0^1\right)$$

$$= -\left(\frac{-e^{-7\pi}}{7\pi} - \frac{1}{49\pi^2}\left(e^{-7\pi}-1\right) + \frac{e^{7\pi}}{7\pi} - \frac{1}{49\pi^2}\left(e^{7\pi}-1\right)\right)$$

$$= \frac{e^{-7\pi}}{7\pi} + \frac{e^{-7\pi}}{49\pi^2} - \frac{1}{49\pi^2} - \frac{e^{7\pi}}{7\pi} + \frac{e^{7\pi}}{49\pi^2} - \frac{1}{49\pi^2}$$

$$= \frac{2}{7\pi}\sin(7\pi) + \frac{2}{49\pi^2}\underbrace{\cos(7\pi)}_{-1} - \frac{2}{49\pi^2} = \boxed{\frac{-4}{49\pi^2}}$$

D ✓

46. Fourierov red: $f(x) = \underbrace{2(1-\cos(5x))}_{\text{parna fja}}$, $x \in \left[-\frac{\pi}{10}, \frac{\pi}{10}\right]$, $T = \frac{1}{5}\pi$

$b_5 = ?$

b_5 je za neparne i ni parne ni neparne $\Rightarrow b_5 = \emptyset$ (A) ✓

47 -||- (B) ✓

48. $f(x) = 7x^{18}$ Fourierov red $S(x)$, $[0, \pi]$ po sinusu. $S(8) = ?$

napomena: $7(8-2\pi)^{18}$ (B) ✓

49. -||- (B) ✓

50. -||- (C) ✓

ne znam točan postupak

51. Fourierov red: $f(x) = -3x\cos(7x)$, $[-\pi, \pi]$, član uz $\cos(28x)$?

$a_n \cdot \cos\left(\frac{2n\pi x}{T}\right) = \longrightarrow \cos(nx)$, $n = 28$

$a_{28} = ? \longrightarrow a_{28} = \frac{1}{\pi}\int_{-\pi}^{\pi} -3x\cos(7x)\cdot\cos(28x)\,dx$

$$a_{28} = \frac{-3}{2\pi}\int_{-\pi}^{\pi} x\left(\cos(35x) + \cos(21x)\right)dx = \frac{-3}{4\pi}\int_{-\pi}^{\pi} x\left(e^{35x} + e^{-35x} + e^{21x} + e^{-21x}\right)dx$$

$$a_{28} = \frac{-3}{2\pi}\left(\frac{x}{35}e^{35x} - \frac{1}{(35)^2}e^{35x} - \frac{x}{35}e^{-35x} - \frac{1}{(35)^2}e^{-35x} + \right.$$

$$\left. + \frac{x}{21}e^{21x} - \frac{1}{(21)^2}e^{21x} - \frac{x}{21}e^{-21x} - \frac{1}{(21)^2}e^{-21x}\right)\Bigg|_{-\pi}^{\pi}$$

$$= \frac{-3}{2\pi}\left(\cancel{\frac{2}{35}\sin(35x)} - \frac{2}{35^2}\cos(35x) + \cancel{\frac{2}{21}\sin(21x)} - \frac{2}{21^2}\cos(21x)\right)\Bigg|_{-\pi}^{\pi}$$

$$= \frac{3}{\pi}\left(\underbrace{\cancel{\frac{1}{35^2}\left((-1) - (-1)\right)}}_{0} + \frac{1}{21}\cdot\emptyset\right) = \emptyset$$ (D) ✓

52. $f(x) = -7x\cos(8x)$, $[-\pi, \pi]$, član uz $\cos(32x) = ?$

$$a_{32} = \frac{-7}{2\pi}\int_{-\pi}^{\pi} x\left(\cos(40x) + \cos(24x)\right)dx = \frac{-7}{4\pi}\int_{-\pi}^{\pi} x\left(e^{40x} + e^{-40x} + e^{24x} + e^{-24x}\right)dx$$

$$= \frac{-7}{4\pi}\left(\cancel{\frac{x}{40}e^{40x}} - \frac{1}{(40)^2}e^{40x} - \cancel{\frac{x}{40}e^{-40x}} - \frac{1}{(40)^2}e^{-40x} + \cancel{\frac{x}{24}e^{24x}} - \frac{1}{(24)^2}e^{24x}\right.$$

$$\left. - \cancel{\frac{x}{24}e^{-24x}} - \frac{1}{(24)^2}e^{-24x}\right)\Bigg|_{-\pi}^{\pi}$$

$$= \frac{-7}{4\pi}\left(-\frac{2}{40^2}\cos(40x) - \frac{2}{24^2}\cos(24x)\right)\Bigg|_{-\pi}^{\pi}$$

$$= \frac{-7}{2\pi}\cdot\emptyset = \emptyset$$ (A) ✓

53. Fourierov integral: $f(x) = g_{[-2\pi, 2\pi]}(t)$, $B(\lambda) = ?$

$$\tilde{f}(x) = \frac{1}{\pi}\int_0^\infty A(\lambda)\cos(\lambda x) + B(\lambda)\sin(\lambda x)\, d\lambda$$

$$B(\lambda) = \int_{-\infty}^{\infty} f(x)\sin(\lambda x)\, dx = \int_{-2\pi}^{2\pi} \sin(\lambda x) = \frac{-1}{\lambda}\cdot\cos(\lambda x)\Big|_{-2\pi}^{2\pi}$$

$$= \frac{-1}{\lambda}\cdot\left(\cos(2\pi\lambda) - \cos(2\pi\lambda)\right) = \underline{\emptyset}$$ A ✓

54. Parna fja $f(x) = e^{-3x}$. Fourierov int:

$$A(\lambda) = \int_0^\infty e^{-3x}\cdot\cos(\lambda x) = \frac{1}{2}\int_0^\infty e^{-3x}\left(e^{\lambda x} + e^{-\lambda x}\right)dx = \underbrace{\frac{1}{2}\int_0^\infty e^{-x(3-\lambda)} + e^{-x(3+\lambda)}\, dx}_{\text{konv. za svaki } x}$$

$$B(\lambda) = \frac{1}{2}\int_0^\infty e^{-x(3-\lambda)} - e^{-x(3+\lambda)}\, dx = \text{konv. za svaki } x$$ B ✓

55. Ako je f parna funkcija zadana s $f(x) = x^2 e^{-2x}$ za $x \geq 0$ onda je $\tilde{f}(\ln(2)) = ?$

Za neprekinute f vrijedi $\tilde{f}(x) = f(x)$

$$\tilde{f}(x) = x^2 e^{-2x} \rightarrow \tilde{f}(\ln(2)) = \ln^2 2\, e^{-2\ln(2)} = \boxed{\ln^2 2\cdot\frac{1}{4}}$$ B ✓

56. f parna: $f(x) = x^5 e^{-2x}$, $\tilde{f}(\ln(2)) = ?$

$$\tilde{f}(x) = f(x) \rightarrow \tilde{f}(\ln(2)) = \boxed{\ln^5 2\cdot\frac{1}{4}}$$ D ✓

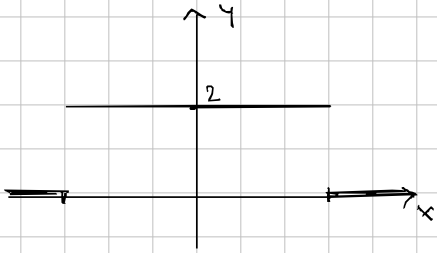
57. Fourierov transformat: $\mathcal{F}\{f(x)\} := \int_{-\infty}^{\infty} e^{-j\lambda x} f(x)\, dx$ A ✓

58. Ako Fourierov transf. $\mathcal{F}\{f'(x)\}$ postoji: $i\lambda\mathcal{F}\{f(x)\}$ D ✓

59. h je neparna fja, F. transf: B čisto imaginarna ✓

60. Fazni spektar C $\arg\hat{f}(\lambda)$ ✓

61. Fourierov transformat: $f(x) = \begin{cases} 2, & |x| \le 3 \\ 0, & |x| > 3 \end{cases}$



$$\hat{f}(\lambda) = 2\int_{-3}^{3} e^{-j\lambda x}\,dx = 2\int_{-3}^{3} \cos(\lambda x) - j\sin(\lambda x)\,dx$$

parna je $\rightarrow j\sin(\lambda x) = \emptyset$

$$2\int_{-3}^{3} \cos\lambda x = 2\cdot\frac{1}{\lambda}\cdot\sin(\lambda x)\Big|_{-3}^{3}$$

$$= \boxed{\frac{4\cdot\sin(3\lambda)}{\lambda}}$$ A

62. $\hat{g}[1,4](\lambda)$ je transf.
gate funkcije $\longrightarrow$ $\hat{g}[1,4](39\pi) = ?$

$$\hat{f}(x) = \int_{-\infty}^{\infty} e^{-j\lambda x} f(x)\,dx \longrightarrow f(x) = \int_{-\infty}^{\infty} e^{j\lambda x}\hat{f}(x)\,dx$$

$$f(x) = \int_{1}^{4} e^{j\lambda x}\,dx = \int_{1}^{4} \cos(\lambda x) + j\sin(\lambda x)\,dx$$

$$= \frac{1}{\lambda}\sin(\lambda x)\Big|_{1}^{4} - \frac{j}{\lambda}\cos(\lambda x)\Big|_{1}^{4} =$$

$$= \frac{1}{\lambda}\left(\sin(4\lambda) - \sin(\lambda) - j\cos(4\lambda) + j\cos(\lambda)\right)$$

$$= \frac{1}{39\pi}\left(\emptyset - \emptyset - j\cdot 1 + j\cdot(-1)\right) = \boxed{\frac{-2j}{39\pi}}$$ B

63. $\hat{g}[1,4](39\pi)$ $\downarrow$ $\frac{1}{39\pi}(-2j) =$ D

64. $\hat{f}$ = Fourierov transf.
F = Laplaceov transf.
gate . fja $g[1,8]$

$$F \rightarrow \int_{0}^{\infty} e^{-st} f(t)\,dt \Rightarrow \int_{0}^{\infty} e^{-j\omega t}\cdot f(t)\,dt = F(j\omega)$$

$$\hat{f} \rightarrow \int_{-\infty}^{\infty} e^{-j\lambda x} f(x)\,dx = \int_{0}^{\infty} e^{-j\lambda x} f(x)\,dx = \hat{f}(\omega)$$ A

65. -||- C